Computer Networks
Fall 2024/25

Exercise-1

Submission in pairs by Thursday, **28-11-2024.** Submission is done by **uploading your work to the course Moodle website**. >>> No late submissions will be accepted.
The name of the submitted file must be Exercise1_username1_username2.[suffix].
For example, Exercise1_israel-israeli_david-davidi.pdf. The first and last names of both students must appear. Only one student should submit the assignment (not both). Please notice that once the grades for an exercise are published, appeals can only be submitted in the following five days. The grader is Mr. Daniel Ohayon, and you may contact him by email at *reichman.computer.networks@gmail.com.*

The exercise covers Chapter 1 from the textbook.

Problem 1 (15 points).

1. A router receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, **one** other packet is **halfway** done being transmitted on this outbound link, and **six** other packets are waiting to be transmitted on this outbound link. Packets are transmitted in order of arrival. Suppose all the packets are **1000 bytes,** and the transmission rate is **1 Mbps** (1M bits per second). What is the queuing delay for the packet? That is, what is the time the packet has to wait in the queue until it arrives at the head of the queue?

2. Consider two hosts, A and B, connected by a single link that supports a transmission rate of **100Kbps**. Suppose the two hosts are separated by **D** meters, and the propagation speed along the link is $\mathbf{2.5 \times 10^8}$ meter/sec. Host A is to send a packet of size **80** bits to host B. What is the distance **D** (in meters) so that the propagation delay equals the transmission delay for this packet? (1 Kbps = 1,000 bps.)

3. Suppose users share a single **5 Mbps** link, and each user requires **0.25 Mbps** when transmitting, but each user transmits only **15 percent** of the time. Is it true that when **circuit switching** networks are used, at most **20** users can be supported simultaneously?

Briefly justify your answers.

Problem 2 (20 points). Use the command ***'tracert'*** (from your cmd) to verify the number of hopes between your home PC (or use https://www.uptrends.com/tools/traceroute if you have problems testing from your PC) to the IP addresses in this question.

Destination IP Address	***IP address of last hope router***	***Number of hopes to destination***	***Name of the destination***
185.228.168.168			
185.228.168.10			
1.1.1.1			
9.9.9.9			
185.228.168.9			
149.112.112.112			
94.140.14.14			
8.8.4.4			
76.76.2.0			

1. (15 points) Please provide screen captures of the results. If you run into problems while performing a route trace, this is probably due to security measures taken by the destination site or your ISP. Any set of measurements is acceptable as long as you show, using screen captures, that you understand the usage of tracert. **Notice:** please embed the screen captions inside your PDF.

2. (5 points) What do all the above nine IP addresses have in common?

Remark: In Lecture 10, I will explain how ***tracert*** works.

Problem 3 (20 points). We wish to send a file of size **200** Mbit from source A to destination B. There are **12** routers on the only existing route between A and B. All the links along the route between A and B support a transmission rate of **5** Mbps, except for the first link, which supports a transmission rate of **10** Mbps, and the second link, which supports a transmission rate of **2** Mbps. Notice that the first link is between A and the first router and not the link between the first two routers! The propagation speed between A and B is $\mathbf{2*10^8}$ meter/sec, and the distance is **1,000,000** km (that is, $\mathbf{10^9}$ meters). We assume the file is divided into **10** equal-size packets for transmission (each of size **20** Mbit), all the routers implement store and forward techniques, and we are disregarding headers overhead. In addition, we assume that when a router is free, it can send a packet as soon as it gets the whole packet. How much time in seconds will it take for the last bit of the file to reach host B? Justify your answer.
Important: Notice that the transmission rate of the first link is **larger** than that of the second link!

Problem 4 (20 points). *This is the only time we will deal with HTML in the course.*
Using only a text editor, compose an HTML page that includes:

1. Use the <title> tag to write: **Computer Networks 2024/25** at the top of the page.
2. The text "**All for one and one for all, united we stand divided we fall**" in blue, at the center of the first line. (This is the motto of the title characters in the book *The Three Musketeers* by the nineteenth-century French author Alexandre Dumas.)
3. One horizontal green line.
4. The text "**The CN 2024 website:**" is in black, followed by a link to this year's course website on Moodle.
5. The text "**The Piazza website:**" is in black, followed by a link to this year's Piazza website.
6. The text "**Our names:**" in brown, followed by your full names.
7. The text "**One of our favorite websites is:**" in green, followed by a single link (in the same line) to your favorite website.
8. The text "**Our favorite image is:**" in red, followed by a single image of your choice.
9. Three images of a dog, a cat, and a bird which are centered, side by side, in the middle of the page.
10. Two images of vegetables centered, side by side, at the middle of the page.

You may want to use a table to align your page better.
Please save a copy of the file with a **.html** extension and open it with your browser.
Please submit (1) a screen capture of your browser (with the opened HTML page), (2) the HTML file, and the images. You are welcome to use the following link to help complete this question: http://www.w3schools.com/html .

Problem 5 (25 points). Enclosed, please find the documents "Wireshark Lab: Getting Started." The goal of this first lab is to introduce you to Wireshark. Download, install, and run the Wireshark software (main page: https://www.wireshark.org/). At the end of the lab, four simple questions will demonstrate that you have been able to get Wireshark up and running. Answer these **4** questions. Whenever possible, justify your answers using screen captures. **Notice: Please do not print your answers as stated in the exercise but embed partial screen captions inside your PDF.**